

|           |                                                                                                 |             |
|-----------|-------------------------------------------------------------------------------------------------|-------------|
|           |                                                                                                 |             |
| 12(a)     | difference between mass of nucleus and mass of (constituent) nucleons                           | <b>M1</b>   |
|           | where nucleons are separated to infinity                                                        | <b>A1</b>   |
| 12(b)(i)  | $E = mc^2$                                                                                      | <b>C1</b>   |
|           | $= 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 / (1.60 \times 10^{-13}) = 934 \text{ MeV}$ | <b>A1</b>   |
| 12(b)(ii) | mass defect = $2 \times (1.007276 + 1.008665) - 4.001506$<br><br>(= 0.030376)                   | <b>B1</b>   |
|           | binding energy per nucleon = $(0.030376 \times 934) / 4 = 7.09 \text{ MeV}$                     | <b>A1</b>   |
| 12(c)     | binding energy per nucleon is much greater                                                      | <b>M1</b>   |
|           | so would require a large amount of energy to separate the nucleons in helium                    | <b>A1</b>   |
|           | <b>or</b>                                                                                       |             |
|           | amount of energy released in forming hydrogen isotopes                                          | <b>(M1)</b> |
|           | is less than energy required to break apart helium nucleus                                      | <b>(A1)</b> |